

BDA Quiz-19 2025

* Indicates required question

1. Email *

2. Roll Number *

3. Password *

Question

4. Here is an example of a tiny Bloom filter. It uses an array of 10 bits and two independent hash functions f and g . We want to test membership in a set S of three elements, so we hash each of the three elements using both f and g , and we set to 1 any bit that any of the three elements is hashed to by either of the hash functions. 1 point

When a new element x arrives, we compute $f(x)$ and $g(x)$, and we say x is in the set S if both $f(x)$ and $g(x)$ are 1. Assume x is not in the set S . What is the probability of a false positive; i.e., the probability of saying that x is in S . Identify this probability, correct to three decimal places, in the list below.

Check all that apply.

- ☐ .220
- ☐ .282
- ☐ .360
- ☐ .780